Free-living cat populations and cat population management in Australia

This briefing brings together the quantitative picture of free-living cat numbers in Australia, how many cats enter Councils, NFP Shelters and Rescues each year, and what the published evidence says about which management measures actually change population size. Every figure is attributed to a named source and time period. Where a figure could not be verified against a primary source it is recorded as a gap rather than stated as fact.

1. How many cats, and of what kind

Australia has no single cat population figure. Cats occupy four quite different ecological niches, and population estimates of each are counted by different methods, on different dates, with very different levels of confidence.

Between three and 10 per cent of Australian adults feed cats they do not consider themselves to own, at an average of 1.5 cats fed daily. These cats sit at the boundary between the owned and unowned populations, and they are the population that community cat programmes reach. Counting them as either owned or feral misrepresents both the number and the intervention that would work.

Category	Estimate	Range	Year
Owned cats	5.3 million	not published	2022
Urban stray (semi-owned plus unowned)	700,000	70,000 to 2.56 million	2016
Feral, natural environments	1.4 million after drought to 5.6 million after rain	1.0 to 2.3 million, and 2.5 to 11 million	2016
Feral, highly modified environments	700,000	not published	2016

Two things in this table matter more than the headline numbers.

The feral cat population varies considerably with environmental conditions

Legge and colleagues estimated the feral population in natural environments fluctuates roughly four-fold between drought years and the years following widespread rain. Adding the 700,000 in highly modified environments such as urban areas, rubbish tips and intensive farms gives a total feral population that moves between about 2.1 and 6.3 million. Any management target expressed as a percentage reduction needs to state which point in that cycle it is measured from.

The urban stray population estimates carry an uncertainty of nearly *forty-fold*

The point estimate of 700,000 sits inside a plausible range of 70,000 to 2.56 million. Expressed per capita that is 29 cats per 1,000 residents, with a range of three to 104.

2. What arrives at the front door

The most complete national picture of cat shelter intake comes from Chua, Rand and Morton's 2023 study, which collected or imputed data from Councils, NFP Shelters and Rescues for 2018-19.

- 179,615 cat admissions nationally, or 7.2 per 1,000 residents.

- Of the admissions that concluded within the year
  - 5% of cats were reclaimed by their guardian
  - 65% were rehomed
  - 28% were euthanased (50,022 cats per annum).

Admission rates vary more than two-fold between jurisdictions, from 4.9 per 1,000 residents in New South Wales to 10.9 in South Australia. That spread reflects differences in what councils do with cats, how many cats are trapped, and how readily people can access an alternative to relinquishment.

Note: Two figures in this dataset need care when quoting. The unduplicated admissions total is 179,615; the sum of intakes across organisation types is 192,584, which double counts cats transferred between organisations. Use admissions for a population picture and intakes for a workload picture.

Other points to note:

- RSPCA Australia publishes the only long, consistent national time series. It covers member societies only, roughly a quarter of national cat admissions, so it is a trend indicator rather than a population count.

- RSPCA cats received fell from 49,166 in 2013-14 to 25,639 in 2024-25, a decline of 48 per cent. This is likely largely due to RSPCA branches pulling back from providing impounding services to local government in multiple states. Euthanasia fell further, from 15,491 to 5,089, a decline of 67 per cent. The euthanasia rate fell from 31.5 per cent to 19.3 per cent by 2023-24 and then moved slightly to 19.8 per cent in 2024-25.

3. What actually changes population size

This is the part of the evidence base most often quoted loosely. CSVs and .pngs covering the data are available in the shared folder

Interpretation notes
- Coverage thresholds
Both sterilisation and lethal removal have a threshold below which they produce no sustained population reduction, and the thresholds are high.

Gunther and colleagues ran a 12-year controlled field experiment and found that neutering more than 70% of the cats in an area was required to reverse population growth, producing an annual reduction of over seven per cent. Hurley and Levy's synthesis puts the sterilisation threshold at 57% to >90% per cent depending on conditions, and the lethal removal threshold at 50% or more of the population annually, a figure independently reached by Andersen, Martin and Roemer's matrix model.

Against those thresholds, euthanasia in Australian shelters and pounds removes about 7% of the urban stray population annually. That is roughly *one seventh* of the lowest published threshold for sustained control. Rand and colleagues attribute the shortfall to three things acting together: the reproductive rate of the remaining cats, immigration into the vacated area, and increased juvenile survival once competition falls.

Why culling below threshold can make things worse

Three lines of field evidence point the same way.

- Lazenby, Mooney and Dickman ran a culling experiment in southern Tasmanian forest and recorded increases of 75 to 211 per cent in the minimum number of cats known to be alive at the culled sites, with numbers declining back to pre-culling levels once culling stopped.
- Palmas and colleagues removed 44 per cent of an identified cat population in 38 days in a semi-isolated setting and recorded full recovery within three months.
- Boone and colleagues' simulation found that episodic culling, repeated whenever the population recovered, reduced final population size only slightly, and that the cumulative number of adult cats that ever lived was *slightly higher than taking no action at all*.

The mechanism is not mysterious. Removal below threshold creates vacant habitat with reduced competition, which increases both immigration and the survival of the kittens that are still being born.

Sterilisation intensity in practice

Where sterilisation has been delivered at real intensity in Australia, intake has fallen substantially.

- The Rosewood Community Cat Program in Ipswich sterilised 308 cats over 3.4 years, an intensity of 27.8 cats per 1,000 residents per year, or 94 per 1,000 cumulatively. Stray cat admissions fell by up to 78 per cent against the 2017 to 2019 baseline, euthanasia fell 85 per cent and cat-related calls to council fell 39 per cent.

- The City of Banyule programme in Victoria worked at 4.1 cats per 1,000 residents per year in three target suburbs over eight years, which is 0.8 per 1,000 measured across the whole city. City-wide cat intake fell 66 per cent and euthanasia fell 82 per cent.

Cotterell, Rand and Scotney recommend planning on ***five to 10 cats per 1,000 residents per year where programmes are micro-targeted to the specific streets and households generating admissions, and 30 to 60 per 1,000 where they are not.*** The difference between those two ranges is the value of targeting. Higher intensity buys faster results rather than a different endpoint.

Note what this means for Banyule at 4.1 per 1,000 in its target suburbs: it sits just below the recommended micro-targeted range and still delivered a two-thirds fall in city-wide intake over eight years. Micro-targeting is what makes a modest absolute number of surgeries move a city-wide figure.

Reproduction, and why thresholds are so high

Free-roaming queens have about 1.4 litters a year with a median of three kittens, and about 75 per cent of those kittens die or disappear before six months, rising towards 90 per cent at high density. ***Combining litters, litter size and kitten survival gives roughly 1.2 kittens surviving to six months per female per year (***figure worked out here rather than a published value and is labelled as such in the file).

That sounds modest. It is enough for ***unmanaged urban populations*** to grow at ***18 to 20 per cent a year,*** because females start breeding at six months and adult mortality is low, at around 5% per six-month interval. Miller and colleagues' elasticity analysis shows why: population growth is far more sensitive to adult survival, at 0.573, than to kitten survival or adult fecundity, at 0.184 each. A cat that survives its first year contributes disproportionately.

Gunther and colleagues also observed compensatory reproduction during the high-intensity phase of their experiment, a 2.25-fold increase in the ratio of kittens to queens. Populations under pressure breed harder, which is precisely why partial coverage does not work and why the thresholds sit as high as they do.

4. Data gaps

There is no national companion animal data standard, and no national collection. The 2023 national study had to request data council by council. Only 60 per cent of pound-operating councils supplied data, and coverage was under a third in South Australia, Queensland, Western Australia and the Northern Territory. Tasmania has no council-operated pounds at all.

The consequences are concrete:

1. The national baseline is 2018-19 and has not been repeated, so there is no way to tell whether national cat intake has moved in seven years.
2. Rescues, which achieve a two per cent euthanasia rate and handle over 41,000 cats a year, are the least visible part of the system in official data.
3. The urban stray population estimate has a range spanning nearly forty-fold, which makes it impossible to state what coverage any given programme achieved.
4. Semi-ownership prevalence rests on a small number of surveys, yet it defines the population that community cat programmes are designed to reach.
5. Household cat ownership prevalence could not be verified against a primary source in this compilation. The owned cat total of 5.3 million reaches this briefing through a secondary citation of the Animal Medicines Australia survey series. The prevalence percentage is recorded as a gap in the master table rather than stated as a figure.

5. What the evidence supports

Read together, the sources point to a fairly narrow set of defensible conclusions.

- Intake reduction and euthanasia reduction have both been achieved at scale in Australia, and in every documented case the mechanism was sterilisation delivered at sufficient intensity in the specific areas generating admissions, provided free or at very low cost to the households involved. Rosewood and Banyule are the two clearest Australian examples.
- Removal-based approaches at the intensity Australia currently applies do not reduce free-living cat numbers, and the field evidence indicates they can be followed by local increases. This is a statement about intensity rather than about principle: at 50 per cent or more annually, removal does reduce populations, but no Australian urban programme operates at anything close to that rate, and shelter-based euthanasia is orders of magnitude short of it.
- Relinquishment and stray admissions concentrate in areas of socioeconomic disadvantage, where sterilisation, microchipping and containment are least affordable. The Rosewood catchment had median individual weekly income of $636 against an Australian average of $805, with 41 per cent of residents renting against 30.6 per cent nationally. Cat containment systems cost between $700 and $2,000. Programmes that treat cost as the binding constraint are working with the grain of the evidence.

Sources

- Chua D, Rand J, Morton J. Stray and Owner-Relinquished Cats in Australia. *Animals*. 2023;13(11):1771. https://doi.org/10.3390/ani13111771

- Legge S, Murphy B, McGregor H, Woinarski J, Augusteyn J, Ballard G, et al. Enumerating a continental-scale threat: How many feral cats are in Australia? *Biological Conservation*. 2017;206:293-303. https://doi.org/10.1016/j.biocon.2016.11.032

- Rand J, M. Saraswathy A, Verrinder J, Paterson MBA. Outcomes of a Community Cat Program Based on Sterilization of Owned, Semi-Owned and Unowned Cats in a Small Rural Town. *Animals*. 2024;14(21):3058. https://doi.org/10.3390/ani14213058

- Cotterell JL, Rand J, Barnes TS, Scotney R. Impact of a Local Government Funded Free Cat Sterilization Program for Owned and Semi-Owned Cats. *Animals*. 2024;14(11):1615. https://doi.org/10.3390/ani14111615

- Cotterell J, Rand J, Scotney R. Urban Cat Management in Australia. *Animals*. 2025;15(8):1083. https://doi.org/10.3390/ani15081083

- Gunther I, Hawlena H, Azriel L, Gibor D, Berke O, Klement E. Reduction of free-roaming cat population requires high-intensity neutering in spatial contiguity to mitigate compensatory effects. *PNAS*. 2022;119(15). https://doi.org/10.1073/pnas.2119000119

- Hurley KF, Levy JK. Rethinking the Animal Shelter's Role in Free-Roaming Cat Management. *Frontiers in Veterinary Science*. 2022;9. https://doi.org/10.3389/fvets.2022.847081

- Lazenby BT, Mooney NJ, Dickman CR. Effects of low-level culling of feral cats in open populations. *Wildlife Research*. 2015;41(5):407-420. https://doi.org/10.1071/WR14030

- RSPCA Australia Annual Statistics. https://www.rspca.org.au/about/annual-statistics/

- Australian Bureau of Statistics, National, state and territory population, June 2025. https://www.abs.gov.au/statistics/people/population/national-state-and-territory-population/jun-2025

Population denominators for all per-1,000-resident figures are ABS estimated resident population at 30 June 2025, except where a source computed its own rate against the population of its own reference year.

Brief for an interactive visualisation: how cat management methods change cat numbers and shelter intake

This pack gives a designer everything needed to build an interactive artifact showing what different cat management methods do, over time, to two things the public and councils both care about: how many free-living cats there are, and how many cats end up in shelters and pounds.

Every default is traceable to an Australian source or to a documented Australian programme. Where a value had to be assumed, it says so.

Files in this pack

- | `viz_scenario_timeseries.csv` | 21 years of output for 11 preset scenarios, indexed to current practice |
- | `viz_model_parameters.csv` | Every parameter with default, plausible range, source and status |
- | `viz_mechanism_comparison.csv` | Side by side comparison of what each method does to the cat, the kittens and the territory |
- | `viz_reference_benchmarks.csv` | Real Australian numbers to anchor the artifact against |
- | `cat_population_model.js` | Dependency-free reference implementation, ES module |
- | `fig7_corrected_mechanism.png` | Static version of the three panels the artifact should show |

1. The one idea the artifact needs to land

Trap-and-remove and sterilise-and-return both reduce cat numbers on paper. They do completely different things to the number of cats entering shelters and pounds, and the reason is mechanical rather than ideological.

Under trap-and-remove, the breeding adult that gets trapped **is** the admission. Trapping is the thing that generates the intake. So the harder a council traps, the more cats arrive at its pound, even as the standing population falls. In the model, doubling today's trapping rate doubles cats entering care immediately, and after 10 years intake is still at 99 per cent of where it started while the population has fallen to 39 per cent.

Under sterilise-and-return, neither the adult nor its future kittens ever enter care. The adult goes back to its caregiver. At five surgeries per 1,000 residents per year, cats entering care fall to 25 per cent of current practice by year 10 while the population falls more slowly, to 65 per cent.

There is a second mechanism, and it is the one that makes removal self-defeating over the longer run. A removed cat vacates the territory and the food resource it was using. That vacancy draws in replacement cats, and arriving cats are sexually entire. A sterilised cat that is returned continues to occupy and hold its territory, so less of the population gets replaced by entire cats. In the proposed model, cats moving in each year at year 10 runs at 18.7 per 10,000 residents under doubled trapping and 21.1 at the highest removal setting, against 15.7 under sterilise-and-return and 11.6 under current practice.

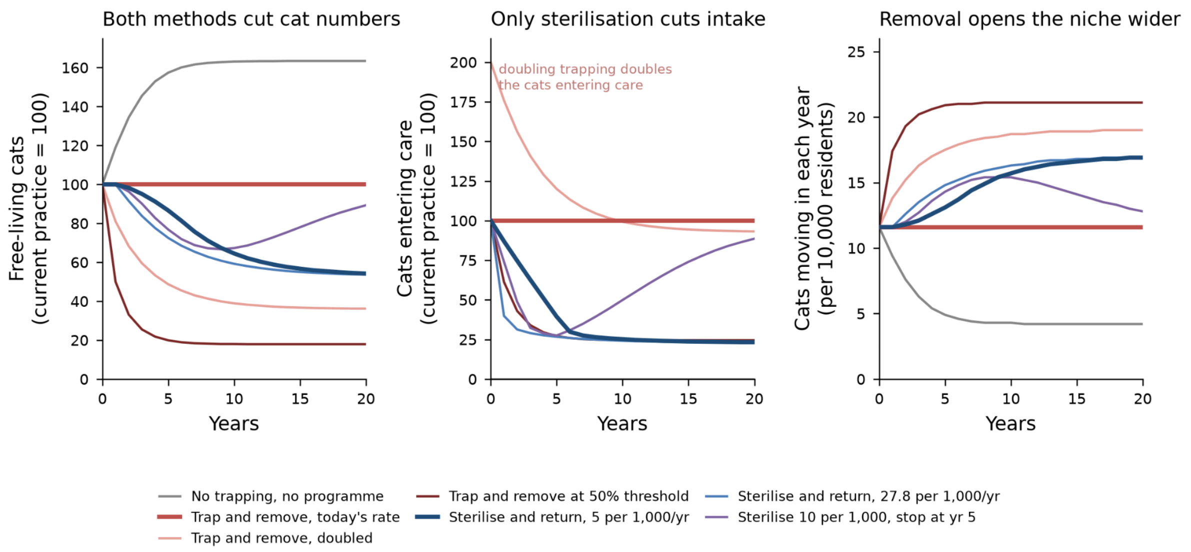
So the honest headline is not "one method works and the other does not". It is: ***removal at sufficient scale can shrink the population, but it cannot reduce the number of cats entering care, because trapping is how they get there. Sterilisation reduces both, and it holds the territory while it does it.***

2. The model

Two compartments, intact and sterilised, annual timestep, expressed per 10,000 residents so it scales to any local government area.

```
N        = I + S                              free-living cats
births  = b_max * max(0, 1 - N/K) * I            only intact cats breed
effPop  = I + (1 - sigma) * S                 cats generating complaints
adultAdm = a_scale * a_ad * effPop               the trapped adult IS the admission
kitAdm   = a_scale * a_kit * births            kittens found or trapped with the queen
removed  = adultAdm * (1 - reclaim)
imm      = m * max(0, K - N)                  vacancy fills with INTACT cats
surg     = min(available intact, x * residents/1000)

I_next = I + births - kitAdm - d*I - removed*(I/effPop) - surg + imm - u_extra*I
S_next = S - d*S - removed*((1-sigma)*S/effPop) + surg - u_extra*S
```



Three structural features carry the whole result, and none of them should be simplified away:

1. **Admissions are an output, not an input.** They are generated by trapping and by births. This is why the intake panel behaves so differently from the population panel.
2. **`sigma` separates the two methods.** A sterilised, microchipped cat with a named caregiver stops generating the complaint that would have had it trapped. Without this term the model cannot reproduce what the two documented Australian programmes actually achieved.
3. **`imm` scales with the emptied niche.** The harder the removal, the faster intact cats arrive. Density dependence in the birth term is not enough on its own; the immigration term is what represents the vacancy argument.

Calibration

Carrying capacity is solved rather than assumed. Taking the observed 29 free-living cats per 1,000 residents as the equilibrium **under today's trapping**, capacity works out at 577 per 10,000 residents, which puts the current population at 50% of capacity. That is a useful internal check, because 50 per cent of capacity is exactly where the published 19 per cent annual growth rate applies.

The admission rate is derived, not guessed. National admissions of 179,615 cats in 2018-19, of which 80 per cent are stray-sourced, set against an urban free-living population of about 735,565, gives 19.5 per cent of the free-living population entering care each year. Note this is much higher than the widely quoted 7 per cent figure, which is the share **killed** annually, not the share admitted.

`sigma` is fitted to reproduce two real programmes: Banyule City Council, where cat impounds fell 66 per cent over eight years at 4.1 surgeries per 1,000 residents, and Rosewood, where stray admissions fell up to 78 per cent over 3.4 years at 27.8 per 1,000. At `sigma` = 0.55 the model returns 72 per cent and 71 per cent respectively. That is a reasonable fit given both programmes also ran education and reclaim initiatives the model does not represent.

3. Parameters, defaults and slider ranges

Full detail in `viz_model_parameters.csv`. The status column matters for how confidently each one can be presented.

Parameter	**Default**	**Range**	**Status**
Free-living cats at start	290 per 10,000 residents	30 to 1,040	Published range
Carrying capacity	577 per 10,000		Calibrated
Adult trapping rate	0.107 per year	0 to 0.5	Derived from national data
Kitten collection rate	0.353 of births	0 to 1	Derived
Kitten share of admissions	0.45	0.30 to 0.65	**Assumed, not published**
Complaint resolution (`sigma`)	0.55	0 to 0.95	Calibrated to two programmes
Surviving kittens per intact cat	0.50 per year	0.35 to 0.70	Calibrated to published growth
Adult mortality	0.10 per year	0.06 to 0.16	Published
Immigration into vacant niche	0.04 of unfilled capacity	0.01 to 0.08	Published, reinterpreted
Reclaim rate	0.05	0.03 to 0.18	Published
Sterilise-and-return rate	5 per 1,000 residents/yr	0 to 30	Published recommendation
Trapping intensity	1.0 (today's rate)	0 to 3.0	Design control
Additional removal	0.0	0 to 0.6	Published threshold at 0.50

Two cautions for whoever builds the sliders.
The surviving-kittens figure of 0.50 is already net of kitten mortality, drawn from 1.4 litters per year at a median of three kittens with about 75 per cent dying before six months. It is not litter size. Anyone tempted to raise it to three or four will produce nonsense.

The kitten share of admissions is genuinely assumed. No Australian source reports an age breakdown of shelter and pound admissions. Banyule reported kitten impounds separately, falling from 85 to 21, but not as a share of total intake. This is worth naming in the artifact as a data gap rather than hiding.

4. Preset scenarios

All 11 are in `viz_scenario_timeseries.csv`, indexed so current Australian practice sits at 100 for both population and intake. Selected results:

Scenario	**Population yr 10**	**Cats entering care yr 10**	**Cats entering care yr 20**
No programme, no trapping	163	0	0
Current practice	100	100	100

Trap and remove, doubled	39	99	93
Trap and remove at the 50 per cent threshold	18	25	25
Sterilise and return, 2 per 1,000/yr	86	61	40
Sterilise and return, 5 per 1,000/yr	65	25	23
Sterilise and return, 10 per 1,000/yr	61	24	23
Sterilise and return, 27.8 per 1,000/yr	59	24	23
Sterilise 10 per 1,000, stop at year 5	67	50	89

The four contrasts worth building the interaction around

1. **Current practice against doubled trapping.** Twice the trapping effort, twice the cats through the pound door on day one, and intake still at 99 per cent of baseline a decade later. This is the counterintuitive result and it should be the first thing a user discovers.

2. **Doubled trapping against 5 per 1,000 sterilisation.** Similar direction of travel on the population, opposite outcomes on intake and on cats moving in.

3. **5 per 1,000 against 10 and 27.8 per 1,000.** Returns flatten out. Five per 1,000 gets most of the intake benefit; 27.8 per 1,000, the Rosewood rate, buys speed rather than a better endpoint. Useful for a council weighing what it can afford.

4. **Sterilise then stop at year 5.** Intake climbs back from 25 to 89 per cent of baseline by year 20 as sterilised cats die and are replaced by entire ones. Programmes are not one-off projects.

One thing to be careful about

The 50 per cent removal threshold scenario (for trap and remove) does reduce both population and intake substantially. It is in the preset list because it is in the literature, *but it should carry a note: removing half of a free-living cat population every year is not something shelters and pounds can deliver, and at current Australian effort the sector removes a small fraction of that. Presenting it as an available option would be misleading.*

5. Real-world anchors

In `viz_reference_benchmarks.csv`. The ones most worth surfacing in the artifact:

- 179,615 cats admitted to Councils, NFP Shelters and Rescues nationally in 2018-19
- 7.1 admissions per 1,000 residents nationally, ranging from 4.9 in New South Wales to 10.9 in South Australia
- 33 per cent of cats entering shelters and municipal facilities in 2018-19 were euthanased
- About 7 per cent of the urban free-living cat population is killed in shelters and pounds each year
- 5 per cent of admitted cats are reclaimed by a guardian nationally
- Strays make up 80 to 100 per cent of admissions to council pounds and 60 to 80 per cent to animal welfare agencies
- Banyule City Council: cat impounds down 66 per cent over eight years, kitten impounds from 85 to 21

- Rosewood: stray admissions down up to 78 per cent over 3.4 years

A user who sets the model to their own local government area population should be able to see their result next to these.

Suggested controls:

- Method selector: trapping intensity and sterilisation rate as two independent sliders, so a user can combine them rather than choosing a side. Real councils run both.
- A residents field, defaulting to something recognisable, that rescales the vertical axes to actual cat numbers and actual surgeries. At five per 1,000 in a 100,000-resident area that is 500 surgeries a year, which is a concrete and checkable ask.
- A "stop the programme" toggle with a year, because the rebound is one of the most decision-relevant behaviours in the model.
- Reveal the parameter table on demand rather than up front, with the status column visible. Anyone who wants to argue with the assumptions should be able to find them in two clicks.

7. What the model does not do

Worth stating plainly in the artifact itself, because the credibility of the thing rests on it.

- It is a single well-mixed population, not a spatial model. Real cat populations are patchy, and micro-targeting sterilisation at the highest density areas is what makes the lower recommended rates achievable. The model cannot show that benefit, so it *understates well-targeted programmes.*
- Owned cats are not represented as a separate compartment. Semi-owned and unowned cats are the free-living population here, and cats moving between owned and free-living status are folded into the immigration and mortality terms.
- The kitten share of admissions is assumed, as noted above.
- Wildlife outcomes are not modelled at all. That is a legitimate concern in the Australian debate and this artifact simply does not speak to it. Better to say so than to imply the question has been answered.
- No spatial or seasonal variation in breeding, and no explicit disease dynamics.
- `sigma` is fitted to two programmes. Two is enough to be suggestive and not enough to be settled.

8. Framing note

Relinquishment and free-living cat numbers both track socioeconomic disadvantage. The areas with the most cats entering pounds are usually the areas where guardians face the highest barriers to desexing: cost, transport, and no nearby low-cost service. That is a systemic pattern rather than a failure of individual care, and the artifact will land better with councils and with the public if it reads that way. A visualisation that shows what a subsidised desexing programme achieves is making an argument about access, not about people who do not care about their cats.

Sources

- Chua, Rand and Morton 2023, *Animals* 13(11):1771. National intake and outcomes. doi:10.3390/ani13111771

- Cotterell, Rand and Scotney 2025, *Animals*. Evidence-based strategies and recommended sterilisation rates. doi:10.3390/ani15081083

- Cotterell et al. 2024, *Animals* 14(21):3058. Banyule City Council programme. doi:10.3390/ani14213058

- Rand et al. 2024, *Animals* 14(11):1615. Rosewood community cat programme. doi:10.3390/ani14111615

- Legge et al. 2017, *Biological Conservation* 206. Continental-scale cat population estimates. doi:10.1016/j.biocon.2016.11.032

- Boone et al. 2019, *Frontiers in Veterinary Science* 6:238. Population model structure and parameters. doi:10.3389/fvets.2019.00238

- Hurley and Levy 2022, *Frontiers in Veterinary Science* 9:847081. Removal thresholds. doi:10.3389/fvets.2022.847081

- Lazenby et al. 2015, *Wildlife Research* 41(5). Population increase following low-level culling. doi:10.1071/WR14030

- Nutter et al. 2004, *JAVMA* 225(9). Reproductive rates and kitten survival. doi:10.2460/javma.2004.225.1399

Two design notes

- The starting population has a published range spanning nearly forty-fold, three to 104 cats per 1,000 residents. Exposing that as a slider is more honest than drawing one confident line.
- And the sterilisation presets at 5, 10 and 27.8 per 1,000 all converge to roughly the same 20-year endpoint. Higher intensity buys speed, not a different destination. If the interface makes the top of the slider look pointless, that needs a note, because speed is what matters to a council trying to cut euthanasia this year.

Honesty constraints

The brief has a section listing things the model cannot do, written so you can put them in an about panel. The two that matter most: there is no spatial structure, yet the 12-year field experiment found sterilisation had to be spatially contiguous to work at all; and there is no targeting quality, yet five per 1,000 delivered to the right streets does what 30 to 60 per 1,000 delivered indiscriminately does. That six-fold efficiency difference is invisible to the model, and if ***you can express targeting as a multiplier on the slider it would be a real improvement on what this model shows.***

I also left compensatory reproduction out. Adding it makes removal look worse, so omitting it is the conservative choice, but the model is therefore optimistic about culling and the about panel should say so.